Capstone Project Submission

Instructions:
1. Please fill in all the required information.
2. Avoid grammatical errors.

Team Member's Name:

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Role: -

1) Data Cleaning: -
- Dealing with null values, duplicate data, and outliers present in our data.

2) Exploratory Data Analysis: -
- Plotting the dependent variable and distributions of dependent and independent variables.
- Checking and visualising the correlation between our dependent and independent variables. visualising the relationship between each pair of our variables.

3) Data Pre-processing & Feature Engineering: -
- Checking for and dealing with multicollinearity present in our dataset.
- Applying the log transform to deal with positively skewed data. Scaling the data and splitting it into train and test sets.

4) Model Implementation: -
- Fitting various models on our data and optimising them via cross-validation.
- Using these models to make predictions on test and train data.

The Models implemented are:
1. Linear Regression
2. Lasso Regression
3. Ridge Regression
4. Elastic Net Regression

5) Data Visualisation: -
- Using several kinds of charts like Line chart, scatter plot, heatmap, pair plot, dist plot, boxplot etc to better visualise data and understand correlation and trends.

6) Model performance comparison:-
- Comparison of all implemented models using various Regression evaluation metrics like Mean absolute error, mean squared error, RMSE, R-squared and adjusted R-squared.

7) Conclusion:-

- Drawing some insights from the data and the predictions made by our various predictive models on unseen (test) data.

Problem statement:-

Yes, Bank is a well-known Indian bank headquartered in Mumbai, India and was founded by Rana Kapoor and Ashok Kapoor in 2004. It offers a wide range of differentiated products for corporate and retail customers through retail banking and asset management services.
Yes, Bank is a publicly traded company listed on the stock market and is therefore subject to the ups and downs of the stock market cycle.

The stock market is driven by speculation. The investors decide on buying or selling shares of a company based on its performance and its reputation. Public opinion has a huge impact on stock market prices. Which is why when the news of a fraud case involving Rana Kapoor broke in 2018, the stock price of Yes Bank went down significantly.

Here we are presented with the stock market price data of Yes Bank, and our job is to try to predict the stock's closing price for the month. This data contains the date, lowest, highest and closing price details.

Our approach is to fit a machine learning model on this past data and try to predict the closing price for new unseen data using the parameters learned during training.
This way, we can get our model to learn the trends present in the data during training and use that information during prediction.

We will apply various Regression Models for this task, such as Linear Regression, Lasso Regression, Ridge Regression, and Elastic Net Regression

Conclusions: -

- Using data visualisation on our target variable, we can clearly see the impact of the 2018 fraud case involving Rana Kapoor, as the stock prices decline dramatically during that period.
- After loading the dataset, we found that there are no null values in our dataset nor any duplicate data.
- There are some outliers in our features; however, this being a very small dataset, dropping those instances will lead to a loss of information.
- We found that the distribution of all our variables is positively skewed. So we performed a log transformation on them.
- There is a high correlation between the dependent and independent variables. This is a signal that our dependent variable is highly dependent on our features and can be predicted accurately from them.
- We found that there is a rather high correlation between our independent variables. This Multicollinearity is, however, unavoidable here as the dataset is very small.
- We implemented several models on our dataset in order to be able to predict the closing price and found that all our models are performing remarkably well, and the Elastic Net regressor is the best performing model with A R2 score value of 0.9938 and scores well on all evaluation metrics.